# POLITECNICO DI TORINO

**Master's Degree in Electronics**

**Master's Degree Thesis**

# Loss analysis of a 4-level,flying capacitor,bridge-less,totem-pole PFC


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# Table of Contents

**1 SPICE model Validation** 3
1.1 Fast Leg GaN FET model validation . . . . . 3
1.1.1 Static charateristics . . . . . 4
1.1.2 Dynamic Charateristics . . . . . 12
1.2 Capacitors model validation . . . . . 15

**List of Tables** 17

**List of Figures** 18

**Bibliography** 19

# Chapter 1

# SPICE model Validation

In order to properly analyze more critical components in terms of losses , is useful to first create a digital model of the device,a validation procedure of the SPICE models of each device has been performed. The following sections describes the procedure and results of this validation procedure.

All simulation results are available [1]

## 1.1 Fast Leg GaN FET model validation

This section describes the behavior of the SPICE model of EPC2304, compared to the EPC datasheet [2]

### 1.1.1 Static charateristics

#### Output I-V charateristics

```
.temp 25

.dc Vds 0 6 0.1 Vgs list 2 3 4 5
```

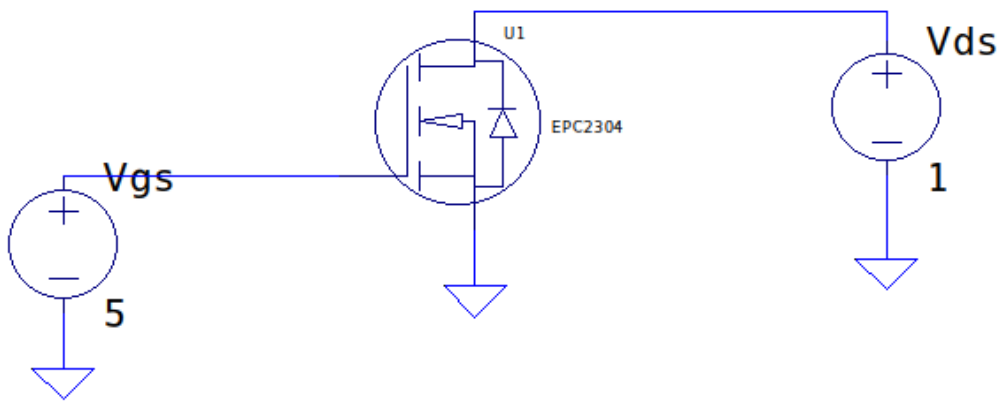


**Figure 1.1:** Schematic for I-V validation

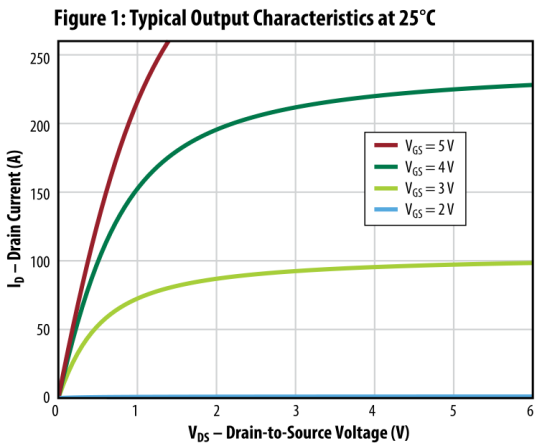


**Figure 1.2:** Datasheet I-V

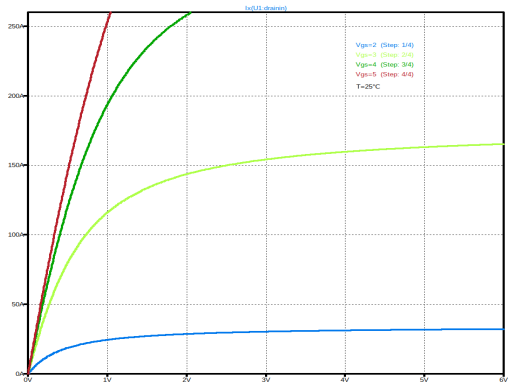


**Figure 1.3:** SPICE I-V

| Vds@Vgs | Ids (datasheet) | Ids (simulation) |
|---|---|---|
| Vds=1V;Vgs=2V | $\approx 0A$ | 25.13A |
| Vds=6V;Vgs=3V | $\approx 100A$ | 163.15A |
| Vds=1V;Vgs=4V | $\approx 150A$ | 187.12A |
| Vds=1V;Vgs=5V | $\approx 215A$ | 253.07A |

**Table 1.1:** I-V data comparision

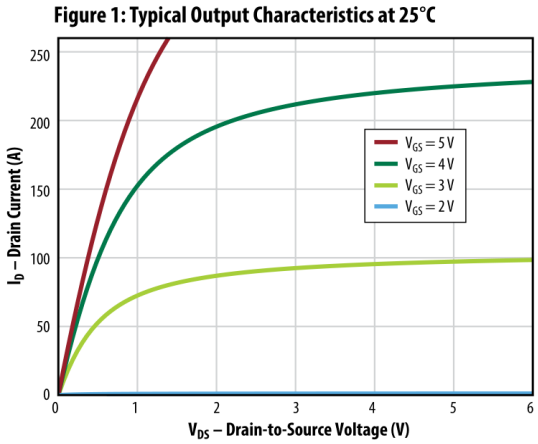


**Figure 1.4:** Datasheet I-V

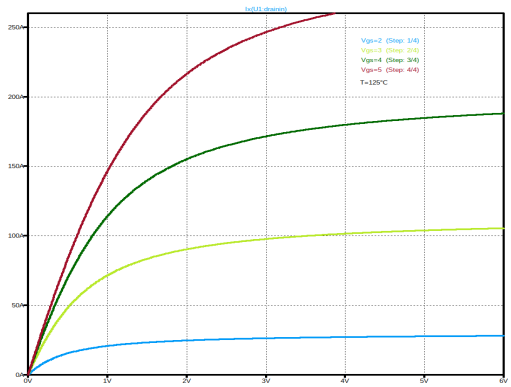


**Figure 1.5:** SPICE I-V @125°C

| Vds@Vgs | Ids (datasheet) | Ids (simulation) |
|---|---|---|
| Vds=1V;Vgs=2V | $\approx 0A$ | 16.41A |
| Vds=6V;Vgs=3V | $\approx 100A$ | 105A |
| Vds=1V;Vgs=4V | $\approx 150A$ | 110.1A |
| Vds=1V;Vgs=5V | $\approx 215A$ | 149.2A |

**Table 1.2:** I-V data comparision

**Trans charateristics**

```
.step temp 25 125 100

.temp 25

.dc Vgs 0.5 5 0.1
```

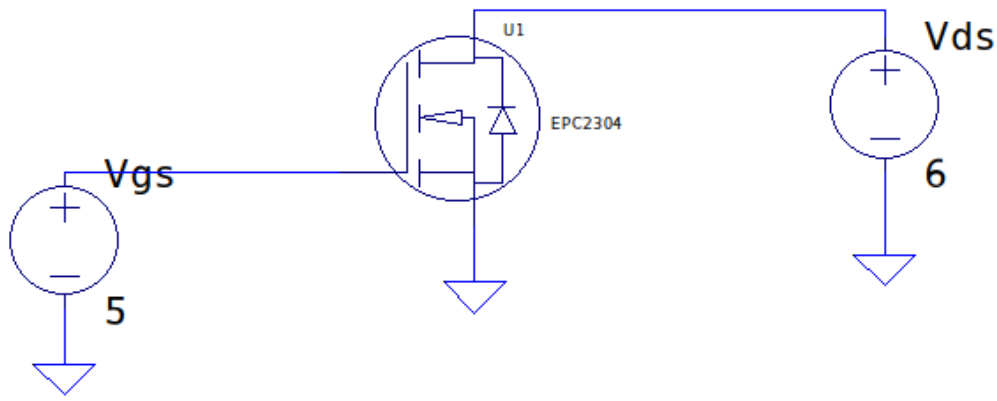


**Figure 1.6:** Schematic for trans charateristics validation

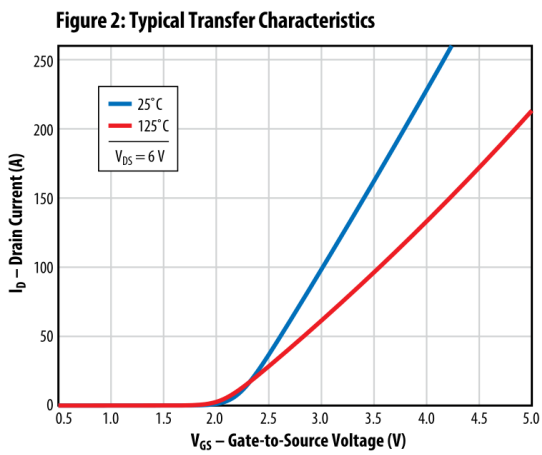


**Figure 1.7:** Datasheet trans charateristics

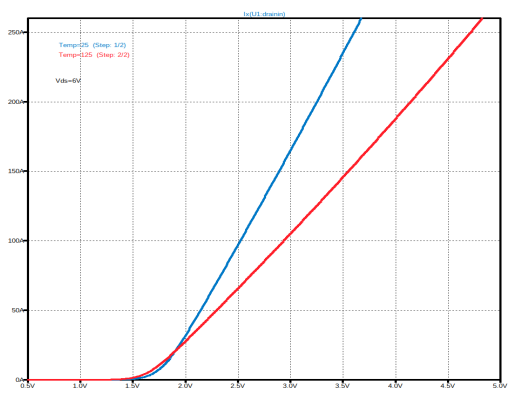

**Figure 1.8:** SPICE trans charateristics

Comparing the 2 it seems like the real device has a Vth of $\approx 2V$,while the simulated one has a Vth of $\approx 1.5V$

| Vgs@T | Ids (datasheet) | Ids (simulation) |
|---|---|---|
| Vgs=2.5V;T=25°C | $\approx 40A$ | 98.58A |
| Vgs=3V;T=25°C | $\approx 100A$ | 163.15A |
| Vgs=3.5;T=125°C | $\approx 100A$ | 148.12A |
| Vgs=4.5V;T=125°C | $\approx 175A$ | 238.07A |

**Table 1.3:** I-V data comparision

### Rdson

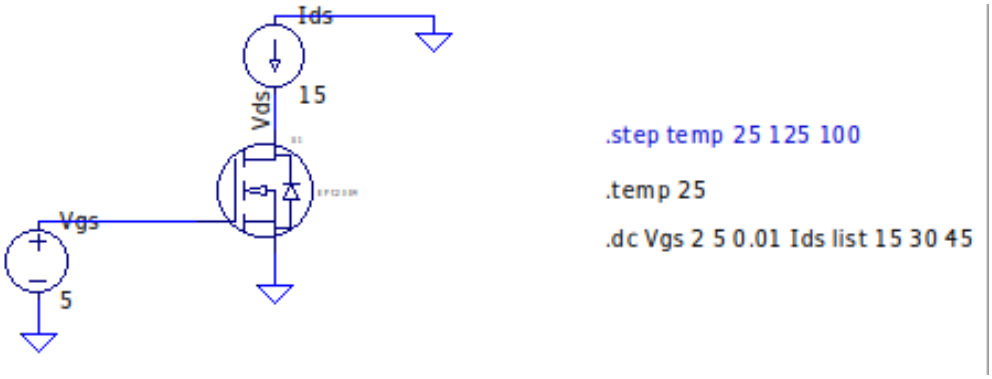


**Figure 1.9:** Schematic for Rdson validation

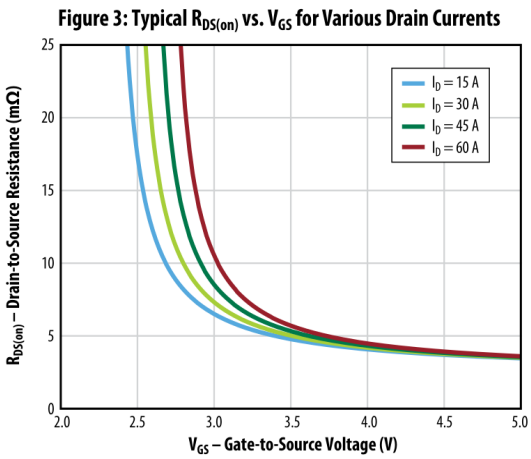


**Figure 1.10:** Datasheet Rdson

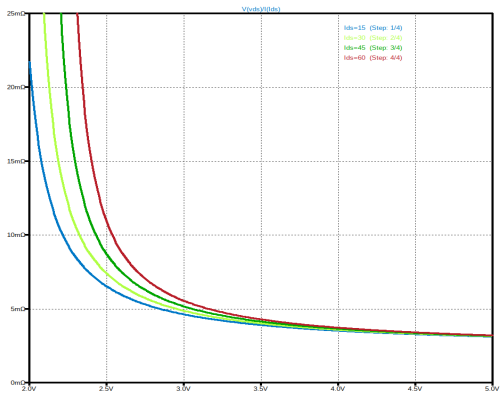

**Figure 1.11:** SPICE Rdson @25°C

| Vgs@Ids | Rds (datasheet) | Rds (simulation) |
|---|---|---|
| Vgs=2.5V;Id=15A | $\approx 17.5m\Omega$ | $6.8m\Omega$ |
| Vgs=3V;Ids=60A | $\approx 10m\Omega$ | $6m\Omega$ |
| Vgs=2V;Ids=15A | $+\infty$ | $22m\Omega$ |
| Vgs=5V | $\approx 4m\Omega$ | $3.11m\Omega$ |

**Table 1.4:** Rds data comparision

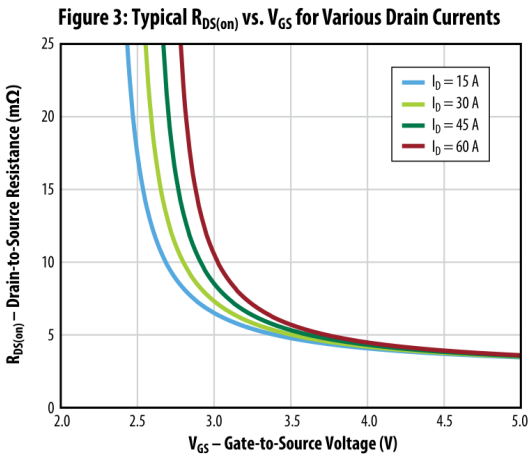


**Figure 1.12:** Datasheet Rdson

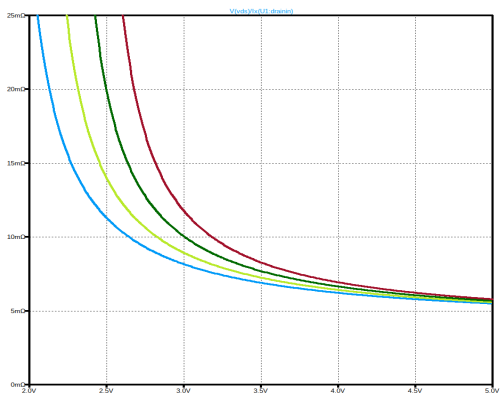

**Figure 1.13:** SPICE Rdson @125°C

| Vgs@Ids | Rds (datasheet) | Rds (simulation) |
|---|---|---|
| Vgs=2.5V;Id=15A | $\approx 17.5m\Omega$ | $11.8m\Omega$ |
| Vgs=3V;Ids=60A | $\approx 10m\Omega$ | $12m\Omega$ |
| Vgs=2V;Ids=15A | $+\infty$ | $30.91m\Omega$ |
| Vgs=5V | $\approx 4m\Omega$ | $5.49m\Omega$ |

**Table 1.5:** Rds data comparision

### Rdson vs tempearture

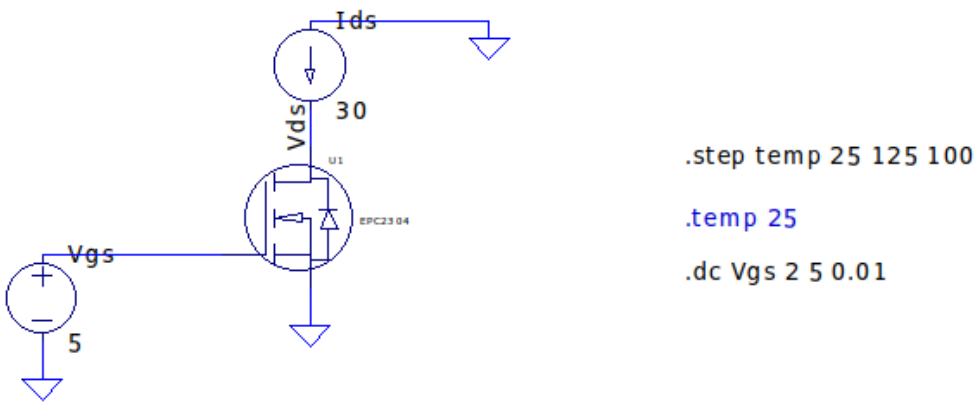


**Figure 1.14:** Schematic for Rdson vs tempearture

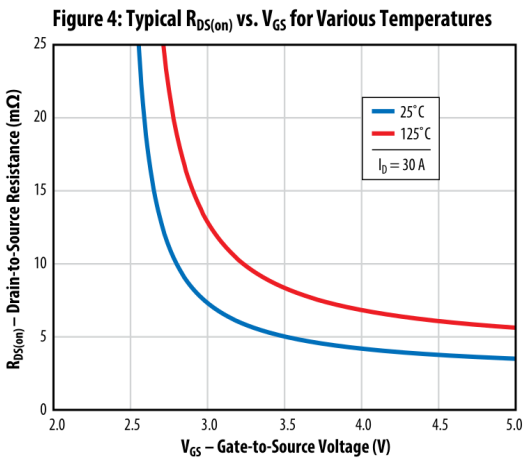


**Figure 1.15:** Datasheet Rdson vs tempearture

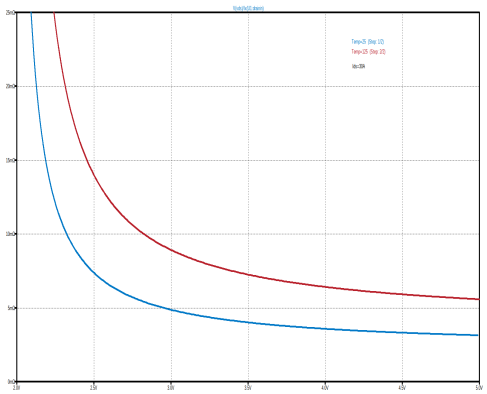

**Figure 1.16:** SPICE Rdson vs tempearture

| Vgs@T | Rds (datasheet) | Rds (simulation) |
|---|---|---|
| Vgs=3.5V;T=25°C | $\approx 5m\Omega$ | $4.6m\Omega$ |
| Vgs=5V;T=125°C | $\approx 5.5m\Omega$ | $5.8m\Omega$ |
| Vgs=2.95V;T=25°C | $\approx 7m\Omega$ | $5m\Omega$ |
| Vgs=2.3V@T=25°C | $+\infty$ | $10m\Omega$ |
| Vgs=2.3V@T=125°C | $+\infty$ | $21.2m\Omega$ |

**Table 1.6:** Rds temperature comparision

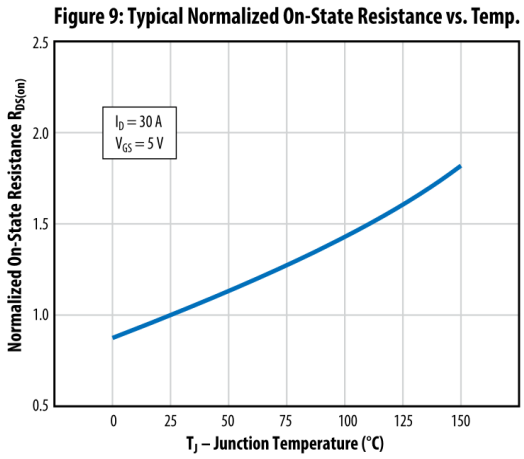


**Figure 1.17:** Datasheet Rdson vs temperature

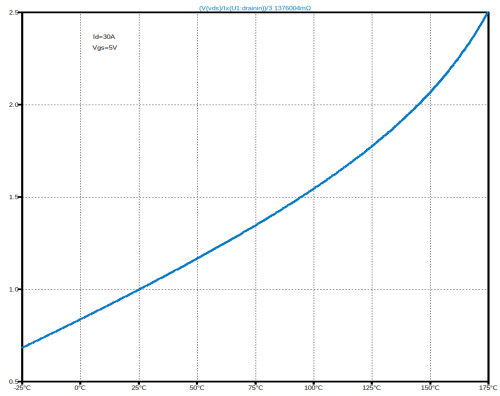


**Figure 1.18:** SPICE Rdson vs temperature

### Reverse body diode

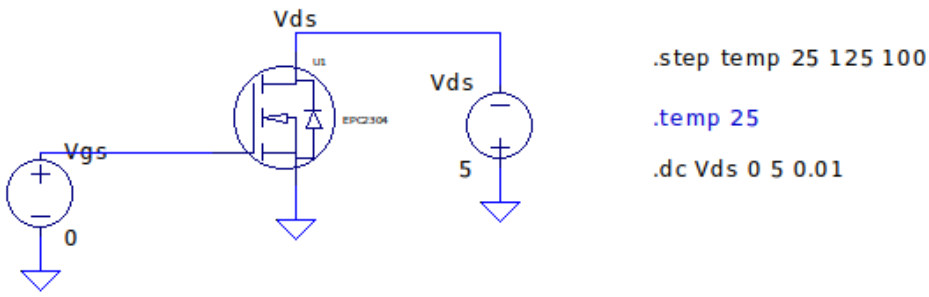


**Figure 1.19:** Schematic for Reverse body diode

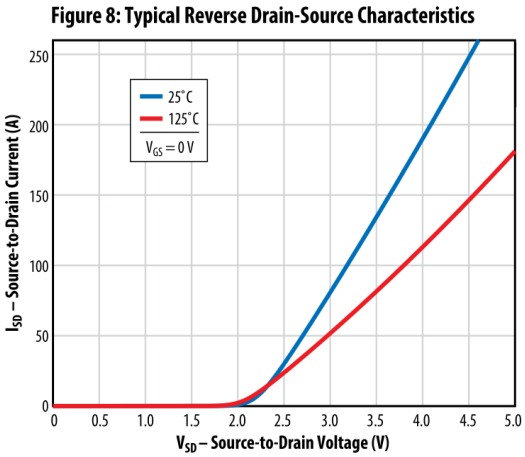


**Figure 1.20:** Datasheet Reverse body diode

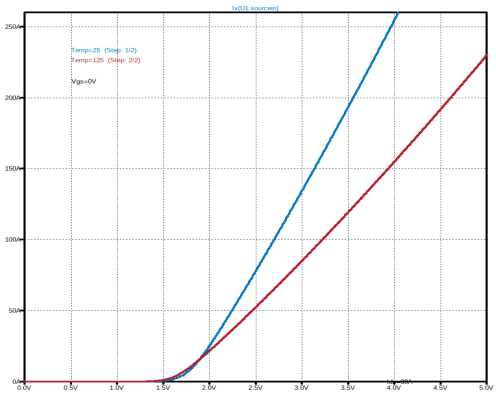

**Figure 1.21:** SPICE Reverse body diode

| Vgs@T | Rds (datasheet) | Rds (simulation) |
|---|---|---|
| Vss=4.5V;T=25°C | $\approx 250A$ | $317A$ |
| Vsd=3V;T=125°C | $\approx 50A$ | $88.7A$ |
| Vsd=4.5V;T=125°C | $\approx 148A$ | $191A$ |

**Table 1.7:** Reverse diode data comparision

### Vth vs temperature

**Figure 1.22:** Schematic for Vth vs temperature

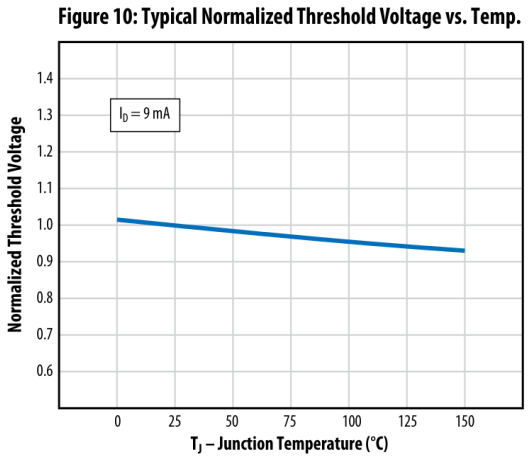


**Figure 1.23:** Datasheet Vth vs temperature

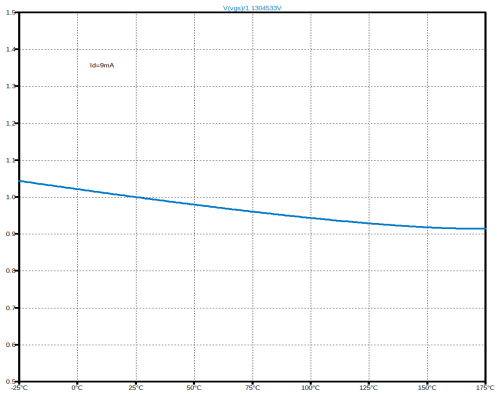

**Figure 1.24:** SPICE Vth vs temperature

### 1.1.2 Dynamic Charateristics

#### Typical switching curves

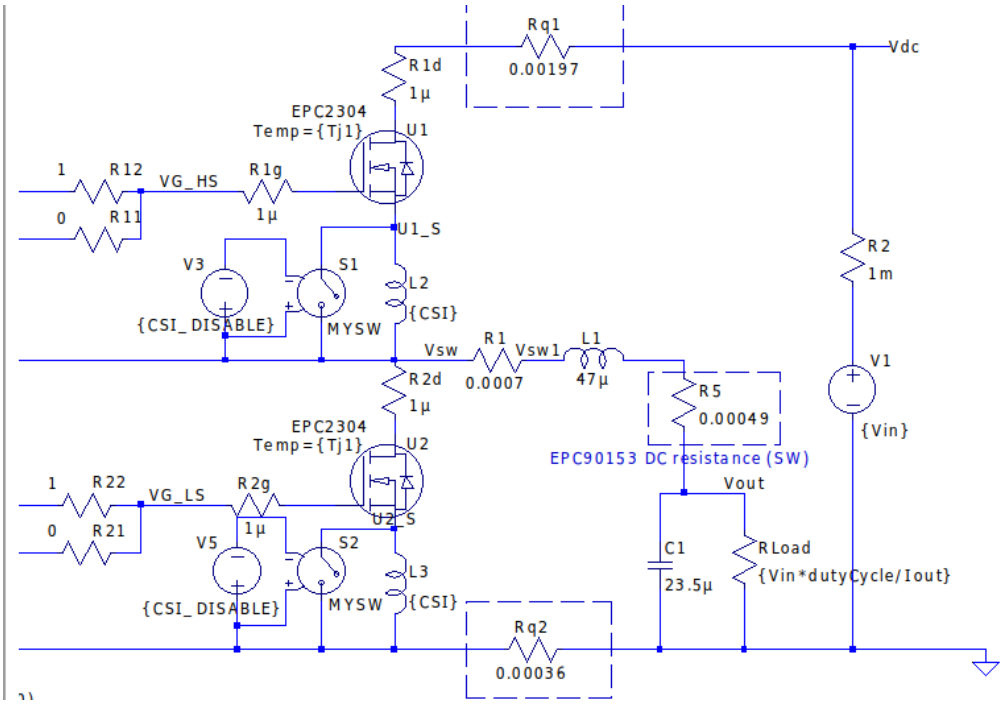


**Figure 1.25:** Schematic for switching behaviour

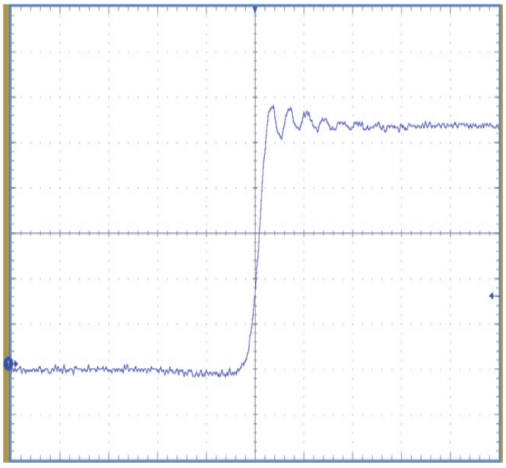

**Figure 1.26:** Datasheet Turn on curve

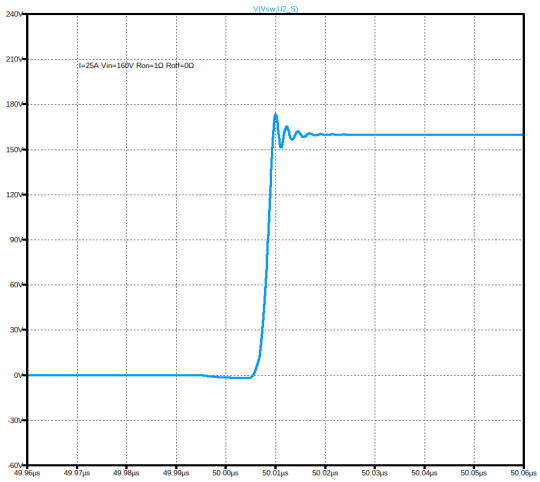


**Figure 1.27:** SPICE Turn on curve

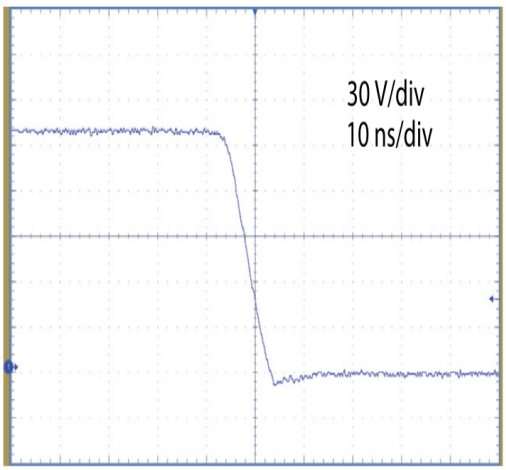


**Figure 1.28:** Datasheet Turn off curve

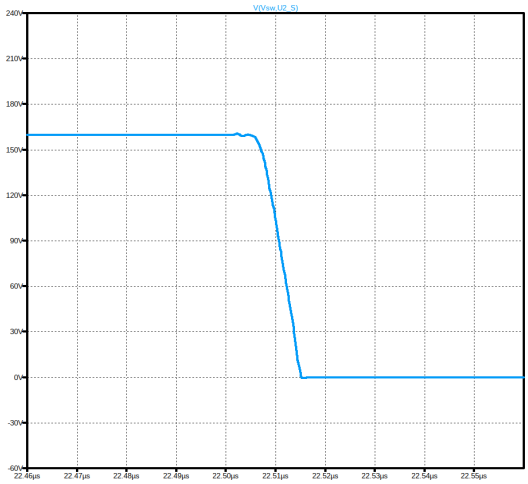


**Figure 1.29:** SPICE Turn off curve

**Typical gate charge**

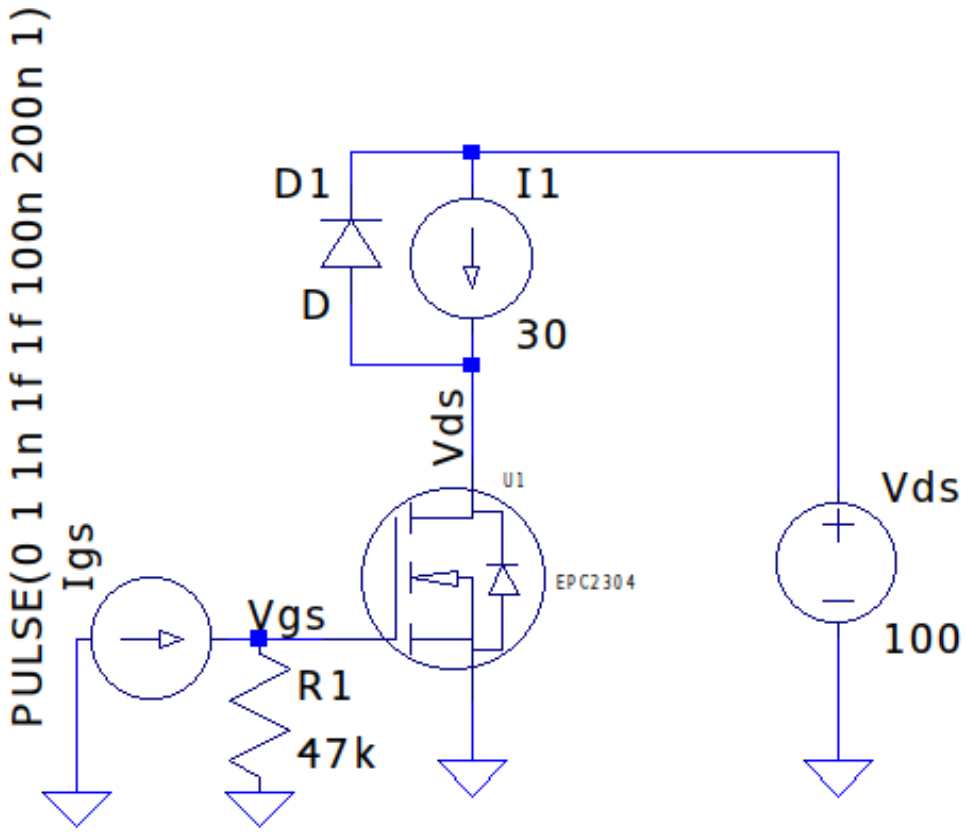


**Figure 1.30:** Schematic for gate charge

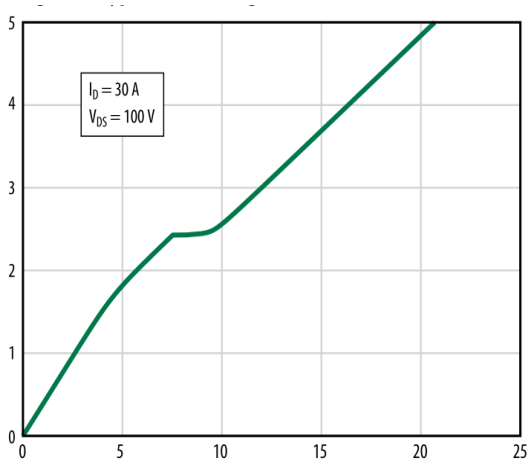


**Figure 1.31:** Datasheet gate charge

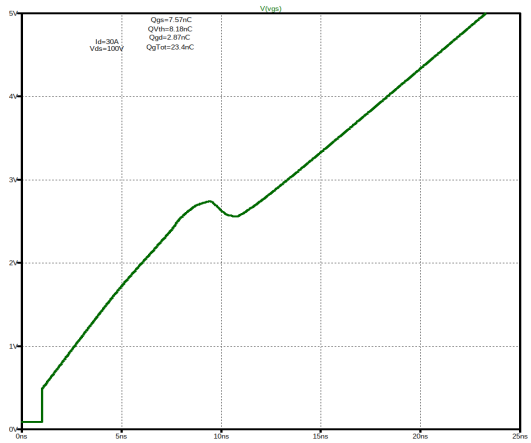


**Figure 1.32:** SPICE gate charge

| Q (datasheet) | Q (simulation) |
|---|---|
| Qgtot=21nC | Qgtot=23.4nC |
| Qgs=7.5nC | Qgs=7.57nC |
| Qgd=2nC | Qgd=2.87nC |
| QVth=5.2nC | QVth=8.18nC |

**Table 1.8:** Gate charge data comparision

## 1.2 Capacitors model validation

This section briefly describes the SPICE model behavior compared to the datasheet described behavior

### Flying capacitor

no particular notes needed,model fits real behavior,no temperature model available [3]

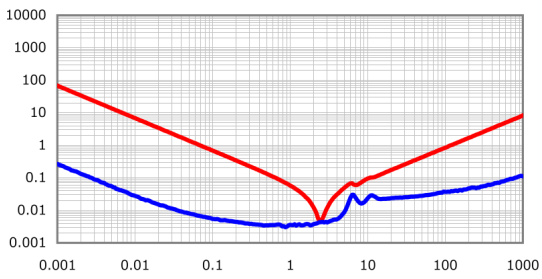


**Figure 1.33:** Datasheet Z

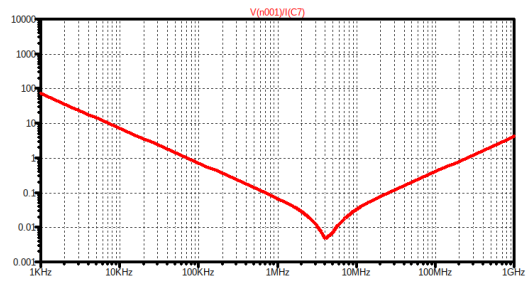


**Figure 1.34:** SPICE Z

### HF decoupling capacitor

no particular notes needed,model fits real behavior,no temperature model available [4]

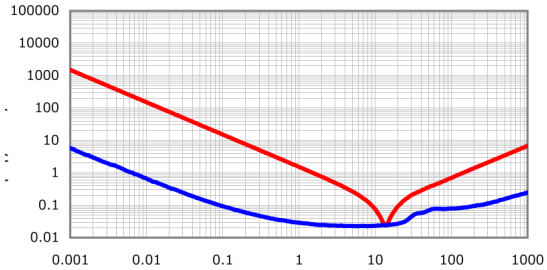


**Figure 1.35:** Datasheet Z

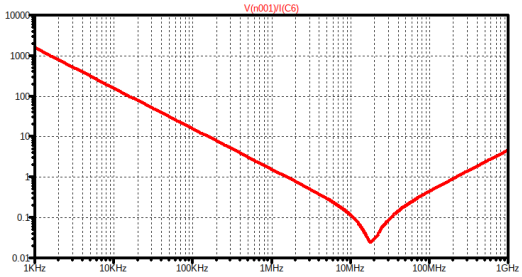


**Figure 1.36:** SPICE Z

# List of Tables

1.1 I-V data comparision . . . . . 5
1.2 I-V data comparision . . . . . 5
1.3 I-V data comparision . . . . . 7
1.4 Rds data comparision . . . . . 8
1.5 Rds data comparision . . . . . 8
1.6 Rds temperature comparision . . . . . 9
1.7 Reverse diode data comparision . . . . . 11
1.8 Gate charge data comparision . . . . . 15

# List of Figures


1.1 Schematic for I-V validation . . . 4
1.2 Datasheet I-V . . . 4
1.3 SPICE I-V . . . 4
1.4 Datasheet I-V . . . 5
1.5 SPICE I-V @125°C . . . 5
1.6 Schematic for trans charateristics validation . . . 6
1.7 Datasheet trans charateristics . . . 6
1.8 SPICE trans charateristics . . . 6
1.9 Schematic for Rdson validation . . . 7
1.10 Datasheet Rdson . . . 7
1.11 SPICE Rdson @25°C . . . 7
1.12 Datasheet Rdson . . . 8
1.13 SPICE Rdson @125°C . . . 8
1.14 Schematic for Rdson vs tempearture . . . 9
1.15 Datasheet Rdson vs tempearture . . . 9
1.16 SPICE Rdson vs tempearture . . . 9
1.17 Datasheet Rdson vs temperature . . . 10
1.18 SPICE Rdson vs temperature . . . 10
1.19 Schematic for Reverse body diode . . . 10
1.20 Datasheet Reverse body diode . . . 11
1.21 SPICE Reverse body diode . . . 11
1.22 Schematic for Vth vs temperature . . . 11
1.23 Datasheet Vth vs temperature . . . 12
1.24 SPICE Vth vs temperature . . . 12
1.25 Schematic for switching behaviour . . . 12
1.26 Datasheet Turn on curve . . . 13
1.27 SPICE Turn on curve . . . 13
1.28 Datasheet Turn off curve . . . 13
1.29 SPICE Turn off curve . . . 13
1.30 Schematic for gate charge . . . 14
1.31 Datasheet gate charge . . . 15
1.32 SPICE gate charge . . . 15
1.33 Datasheet Z . . . 15
1.34 SPICE Z . . . 15
1.35 Datasheet Z . . . 16
1.36 SPICE Z . . . 16

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